ISE 5406: Optimization II

Lecture 7

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Reading for this lecture

- Sections 3.1 and 3.3 of Bazaraa et al. (2006)
- Section 3.1 of Boyd and Vandenberghe

Outline

The Gradient of a Multivariate Function

Definition: Let $S \subseteq \mathbb{R}^n$ be nonempty. A function $f : S \to \mathbb{R}$ is said to be differentiable at $\bar{\mathbf{x}} \in \text{int}(S)$ if there exists a gradient vector $\nabla f(\bar{\mathbf{x}})$ and a function $\alpha : \mathbb{R}^n \to \mathbb{R}$ such that

$$f(\mathbf{x}) = \underbrace{f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}})}_{\substack{\text{First-order Taylor series} \\ \text{approximation of } f \text{ near } \bar{\mathbf{x}}}} + \|\mathbf{x} - \bar{\mathbf{x}}\| \, \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) \qquad \text{for each } \mathbf{x} \in S,$$

where $\lim_{\mathbf{x} \to \bar{\mathbf{x}}} \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) = 0$.

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where $\lim_{\mathbf{x} \to \bar{\mathbf{x}}} \alpha(\bar{\mathbf{x}}; \mathbf{x} - \bar{\mathbf{x}}) = 0$.

Note: If f is differentiable at $\bar{\mathbf{x}}$, there is a *unique* gradient vector given by

$$\nabla f(\bar{\mathbf{x}}) = \left(\frac{\partial f(\bar{\mathbf{x}})}{\partial x_1}, \ldots, \frac{\partial f(\bar{\mathbf{x}})}{\partial x_n} \right) \in \mathbb{R}^n$$

Differentiable Convex Functions

Lemma: Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be a convex function. Suppose f is differentiable at $\bar{\mathbf{x}} \in \text{int}(S)$. Then f has a unique subgradient $\nabla f(\bar{\mathbf{x}})$ at $\bar{\mathbf{x}}$, i.e., $\partial f(\bar{\mathbf{x}}) = \{\nabla f(\bar{\mathbf{x}})\}$.

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Theorem

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be differentiable on S. Then f is convex if and only if for any $\bar{\mathbf{x}} \in S$:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for each } \mathbf{x} \in S.$$

- Affine function $f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$ bounds f from below, which can be used to construct polyhedral outer-approximations of f.

Directional Derivative of Convex Functions

Definition

Let $S \subseteq \mathbb{R}^n$ be a nonempty set and $f : S \to \mathbb{R}$. Let $\bar{\mathbf{x}} \in S$ and $\mathbf{d} \in \mathbb{R}^n$ be a nonzero vector such that $\bar{\mathbf{x}} + \lambda\mathbf{d} \in S$ for $\lambda > 0$ sufficiently small. The directional derivative of f at $\bar{\mathbf{x}}$ along the vector $\mathbf{d}$ is defined as

$$f'(\bar{\mathbf{x}}; \mathbf{d}) = \lim_{\lambda \to 0^+} \frac{f(\bar{\mathbf{x}} + \lambda\mathbf{d}) - f(\bar{\mathbf{x}})}{\lambda}$$

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Lemma: Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Consider any point $\bar{\mathbf{x}} \in \mathbb{R}^n$ and a nonzero direction $\mathbf{d} \in \mathbb{R}^n$. Then the directional derivative $f'(\bar{\mathbf{x}}; \mathbf{d})$ of f at $\bar{\mathbf{x}}$ in the direction $\mathbf{d}$ exists.

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$$f'(\bar{\mathbf{x}}; \mathbf{d}) = \nabla f(\bar{\mathbf{x}})^T \mathbf{d}.$$

Taylor Series Approximations for Multivariate Functions

Taylor Series: Given a smooth univariate function $f : \mathbb{R} \to \mathbb{R}$, we can approximate it with a polynomial around $\bar{x} = a$ as follows:

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f'''(a)(x-a)^3 + \dots$$

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- Second-order approximation:

$$f(\mathbf{x}) \approx f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) + \frac{1}{2}(\mathbf{x} - \bar{\mathbf{x}})^T H(\bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})$$

Hessian matrix $H(\bar{\mathbf{x}}) = \nabla^2 f(\bar{\mathbf{x}})$ is defined by $H_{ij}(\bar{\mathbf{x}}) = \frac{\partial^2 f(\bar{\mathbf{x}})}{\partial x_i \partial x_j}$, $\forall i, j \in [n]$

Twice Differentiable Functions

Definition: Let S be a nonempty set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$. Then f is said to be twice differentiable at $\bar{\mathbf{x}} \in \text{int}(S)$ if there exists

- a gradient vector $\nabla f(\bar{\mathbf{x}})$,
- an $n \times n$ symmetric Hessian matrix $H(\bar{\mathbf{x}}) = \nabla^2 f(\bar{\mathbf{x}})$, and
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such that for each $\mathbf{x} \in S$:

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Examples

1. $f(\mathbf{x}) = 4x_1 + 2x_2 - 7x_1^2 - 3x_2^2 + 5x_1x_2$ at $\bar{\mathbf{x}} = \mathbf{0}$.
2. $f(\mathbf{x}) = e^{2x_1 + 3x_2}$ at $\bar{\mathbf{x}} = (2, 1)$.

Positive Semidefinite and Negative Semidefinite Matrices

Definitions: A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is said to be

- **Positive Semidefinite (PSD):** if $\mathbf{z}^T A \mathbf{z} \geq 0$ for all $\mathbf{z} \in \mathbb{R}^n$.

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Examples

1. $A = 2 \times 2$ identity matrix
2. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

Checking if a Matrix is Positive Semidefinite

Lemma: Consider a symmetric matrix $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \in \mathbb{R}^{2\times 2}$. Then A is positive semidefinite if and only if $a \geq 0$, $c \geq 0$, and $ac - b^2 \geq 0$. It is positive definite if and only if all of these inequalities are strict.

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In practice: check positive definiteness of A by computing Cholesky factorization.

Twice Differentiable Convex Functions

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Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be twice differentiable on S. Then f is convex <u>if and only if</u> the Hessian matrix $H(\bar{\mathbf{x}}) = \nabla^2 f(\bar{\mathbf{x}})$ is positive semidefinite at each $\bar{\mathbf{x}} \in S$.

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Note: If the function is quadratic, the Hessian matrix is independent of the point under consideration. Hence, checking its convexity reduces to checking the positive semidefiniteness of a constant matrix in this case.

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Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be twice differentiable on S. If the Hessian matrix is positive definite at each point in S, then f is strictly convex.

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Recap: Some Examples of Convex and Concave Functions

Some examples of convex functions:

- Exponential function: $\exp(x)$ on $X = \mathbb{R}$. (Strictly convex)
- Power functions: x^p on $X = (0, +\infty)$ for $p \geq 1$ or $p \leq 0$.
- Negative entropy: $x \ln(x)$ on $X = (0, +\infty)$. (Strictly convex)
- Quadratic functions: $\mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ on $X = \mathbb{R}^n$ for any $A \in \mathcal{S}_+^n$, $\mathbf{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$. (Strictly convex if A is positive definite)

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Some examples of concave functions:

- Power functions: x^p on $X = (0, +\infty)$ for $p \in [0, 1]$.
- Geometric mean: $\left(\prod_{i=1}^n x_i\right)^{1/n}$ on $X = \mathbb{R}_{++}^n$.
- Logarithm: $\ln(x)$ on $X = (0, +\infty)$. (Strictly concave)
- Quadratic functions: $\mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ on $X = \mathbb{R}^n$ for $-A \in \mathcal{S}_+^n$, $\mathbf{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$. (Strictly concave if A is negative definite)

Definition: Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function and S be a nonempty convex set in $\mathbb{R}^n$. Then a convex optimization problem is defined by

$$\begin{aligned} &\min \quad f(\mathbf{x}) \\ &\text{s.t.} \quad \mathbf{x} \in S. \end{aligned} \tag{1}$$

Theorem

Suppose that $\bar{\mathbf{x}} \in S$ is a local optimal solution to Problem (**??**).

1. Then $\bar{\mathbf{x}}$ is a global optimal solution.
2. If either $\bar{\mathbf{x}}$ is a strict local minimum or f is strictly convex, then $\bar{\mathbf{x}}$ is the unique global optimal solution (it is also a strong local minimum).

Recap: Necessary & Sufficient Conditions for a Global Min.

Theorem

Point $\bar{\mathbf{x}} \in S$ is an optimal solution to Problem (**??**) if and only if f has a subgradient $\boldsymbol{\xi}$ at $\bar{\mathbf{x}}$ such that $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$.

Corollary 1: Under the assumptions of the above theorem, if S is an open set (for example, $S = \mathbb{R}^n$), $\bar{\mathbf{x}}$ is an optimal solution to the problem if and only if there exists a zero subgradient of f at $\bar{\mathbf{x}}$.

Corollary 2: Assume that f is differentiable. Then $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$. Furthermore, if S is open, $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}}) = \mathbf{0}$.

Examples

- **Solve**

$$\begin{aligned} \min\ & x_1 + x_1 x_2 + x_2^2 \\ \text{s.t.}\ & (x_1, x_2) \in \mathbb{R}^2. \end{aligned}$$

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